

Here's a concise Kubernetes (kubectl) cheat sheet you can use for daily practice and troubleshooting. It covers cluster, pod, deployment, service, and config management — the most common commands you'll need.

Core Cluster Commands

Cluster Info

```
bash
kubectl cluster-info
kubectl get nodes
kubectl describe node <node-name>
```

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Namespaces

```
bash
kubectl get namespaces
kubectl create namespace dev
kubectl delete namespace dev
```

•

Pod Management

Create Pod from YAML

```
bash
kubectl apply -f pod.yaml
```

•

View Pods

```
bash
kubectl get pods
kubectl get pods -o wide
kubectl describe pod <pod-name>
```

•

Delete Pod

```
bash
kubectl delete pod <pod-name>
```

-

Deployments & ReplicaSets

Create Deployment

bash

```
kubectl create deployment nginx --image=nginx
```

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Scale Deployment

bash

```
kubectl scale deployment nginx --replicas=3
```

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Update Image

bash

```
kubectl set image deployment/nginx nginx=nginx:1.25
```

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View Rollout Status

bash

```
kubectl rollout status deployment/nginx
```

-

Services & Networking

Expose Deployment as Service

bash

```
kubectl expose deployment nginx --port=80 --type=NodePort
```

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Get Services

bash

```
kubectl get svc
```

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Describe Service

bash

```
kubectl describe svc nginx
```

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ConfigMaps & Secrets

Create ConfigMap

bash

```
kubectl create configmap app-config --from-literal=ENV=prod
```

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Create Secret

bash

```
kubectl create secret generic db-secret --from-literal=PASSWORD=pass123
```

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View Configs

bash

```
kubectl get configmaps
```

```
kubectl get secrets
```

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Monitoring & Debugging

Logs

bash

```
kubectl logs <pod-name>
```

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Exec into Pod

bash

```
kubectl exec -it <pod-name> -- /bin/bash
```

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Resource Usage

bash

```
kubectl top nodes
```

```
kubectl top pods
```

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Quick Tips

- Use `-n <namespace>` to target specific namespaces.
- Use `-A` for all namespaces.
- Alias `kubectl` to `k` for faster typing:

bash

```
alias k=kubectl
```